

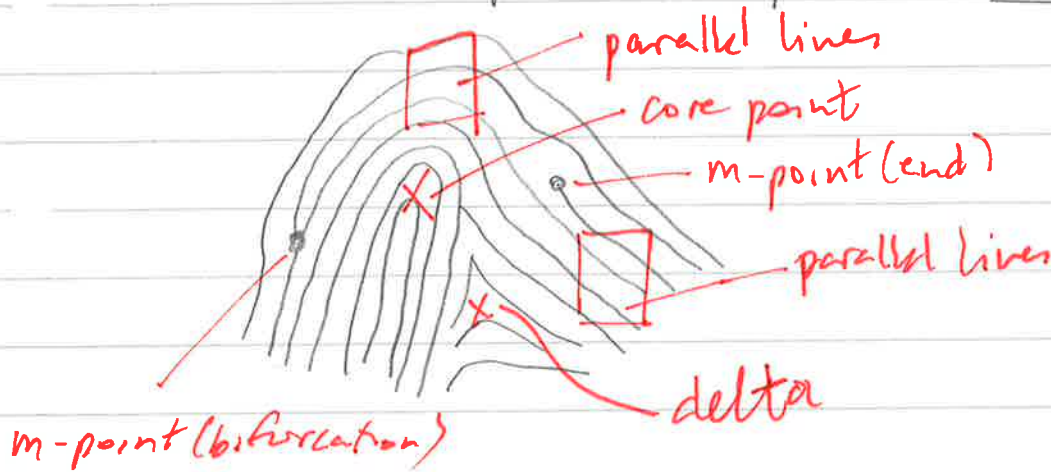
Bionic Recognition FINGERPRINT RECOGNITION

lab exercise 4

* DETECTION OF PATTERNS IN FINGERPRINTS by LINEAR FILTERING *

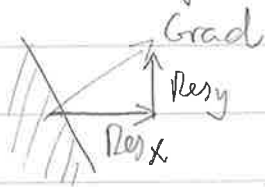
- fingerprint images are composed of several patterns:

- parallel lines
- core / delta points : one or two of each per image
- minutiae points (m-points) : dozens per image



- Here we will see how to detect these patterns by linear filtering

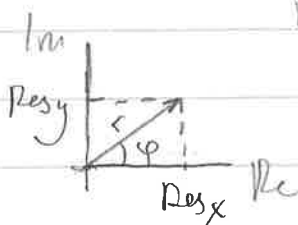
- Remember gradient from linear filtering exercise (p. 8):



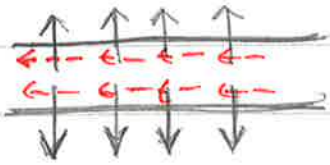
Res_x = change in x-direction
 Res_y = change in y-direction

- let's express gradient as a complex number as follows:

$$\text{Gradient} = \underbrace{Res_x}_{\text{Real}} + i \underbrace{Res_y}_{\text{Imaginary}} = \underbrace{r}_{\text{magnitude}} e^{i\phi}_{\text{angle}}$$



(2)

A) DETECTION of parallel LINES

$$90^\circ \xrightarrow{2x} 180^\circ$$

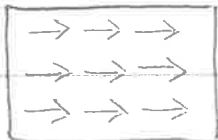
$$270^\circ \xrightarrow{2x} 540^\circ - 360^\circ = 180^\circ$$

double angle:
opposite lines
become the same

- When computing gradient (edges), we double the angle

$\uparrow = r \cdot e^{i\phi}$
compute gradient

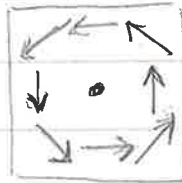
$\leftarrow = r \cdot e^{i2\phi}$
double angle



filter hp
(detector of
parallel lines)



fp (region of parallel
lines)



fc (core point)

Gradient image + double angle

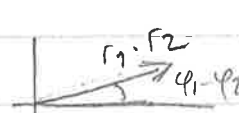
- Linear filtering: scalar product $\langle f, h \rangle$ calculated for each position in the double angle image

- How do we multiply "two arrows"?

$$r_1 \uparrow \phi_1 = r_1 e^{i\phi_1}$$

$$r_2 \nearrow \phi_2 = r_2 e^{i\phi_2}$$

$$r_1 e^{i\phi_1} \cdot r_2 e^{-i\phi_2} = r_1 r_2 e^{i(\phi_1 - \phi_2)}$$

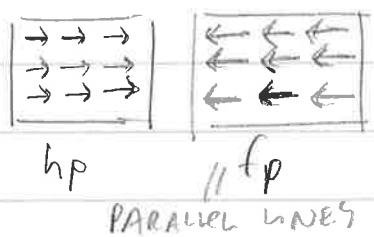


With complex vectors, scalar product is computed by conjugating the second vector

$$\langle f, h \rangle = \sum_i f_i h_i^*$$

Magnitude: multiplied $r_1 \cdot r_2$
Angle: difference $\phi_1 - \phi_2$

- Multiplication with filter h_p : $r=1$
 $\varphi=180^\circ$
 $r=1$
 $\varphi=0$

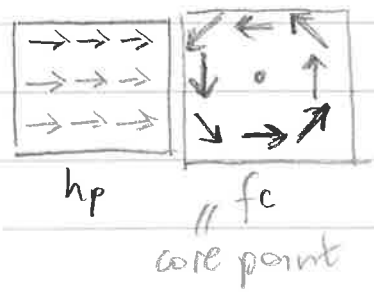


$$\langle f_p, h_p \rangle = \leftarrow \bullet \rightarrow + \leftarrow \bullet \rightarrow + \dots + \leftarrow \bullet \rightarrow$$

$$= \leftarrow + \leftarrow + \dots + \leftarrow$$

$$= \leftarrow$$

result = Long vector with same direction of the parallel lines



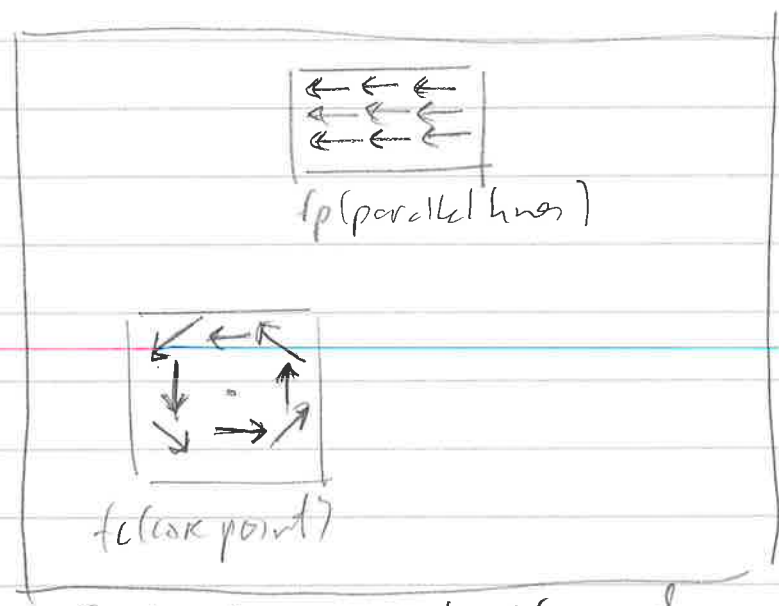
$$\langle f_c, h_p \rangle = \uparrow \bullet \rightarrow + \uparrow \bullet \rightarrow + \dots + \uparrow \bullet \rightarrow$$

$$= \text{circular arrows} = \text{"zero" vector}$$

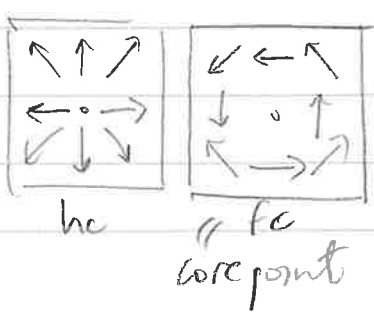
result = no vector ("zero") $\Rightarrow$ no parallel lines in this point

B) DETECTION OF CORE-POINT PATTERN

filter h_c
 (detector of core points)



Gradient image + double angle



$$\langle f_c, h_c \rangle = \uparrow \bullet \rightarrow + \uparrow \bullet \nearrow + \dots + \nearrow \bullet \downarrow$$

$$= \begin{matrix} \uparrow \\ \uparrow \\ \uparrow \end{matrix}$$

result = Long vector with direction = 90°

(difference of angles between arrows is 90°)

- Remember:

LS = "long arrow" if lines are parallel

PS = "long arrow" in core-point patterns

- Then m-points are detected by checking:

$$M = |PS| (1 - |LS|)$$

→ $|\cdot|$ = magnitude of the vector

→ Let's assume that $|PS| \in [0, 1]$
 $|LS| \in [0, 1]$

≥ 1 core-patterns
 ≥ 0 otherwise

≥ 1 parallel lines
 ≥ 0 core-patterns

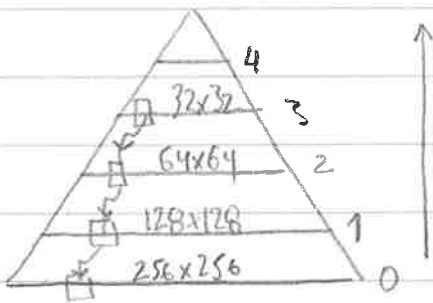
$M \in [0, 1]$: takes high value in positions where there are m-points

→ m-points are like "core-patterns" (high $|PS|$)

→ $(1 - |LS|)$ is used to reinforce the fact that at m-points, the existence of parallel lines is "broken"

(m-points are "small core-patterns", thus $|PS|$ may not be sufficient by itself)

D) PYRAMID DETECTION OF CORE/M-POINTS



→ Core/delta points are "supported" by the entire image (all image lines are arranged around them)

↓
 can be detected up in the scale, where other details (m-points) are removed

1) Low-pass filter (remove details)

2) Half the size

→ m-points are local details of the image, must be detected at the highest resolution (scale 0)

• In scale 0: we have all the details (m-points)

• In higher scales (1, 2, 3, 4) details are removed, only global details remain (like delta and core)

- So the detection procedure is

x core/delta points are available in all scales of the pyramid, thus find them in a higher level (e.g. 3)

x then go down to lower scales and refine the position of the detected core/delta by looking around the position from the previous scale. Go until the scale 0 (highest resolution)

x m-points are only available at lower scales, detect them at highest resolution (scale 0)



→ m-points are difficult to detect anyway if quality of the fingerprint is bad

→ then it may be necessary to enhance the image before the gradient is computed (will see in the lab exercise)